

Mathematics - Hall and Knight

Time Allowed : 1 hour _____ **Maximum Marks : 100**

Please read the instructions carefully. You will be allotted 5 minutes specifically for this purpose.

Instructions

A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

B. Question paper format and Marking Scheme :

1. The paper contains 10 questions. Each question in this section contains 10 marks.

- Q1:** There are 10 steamers plying between Liverpool and Dublin; in how many ways can a man go from Liverpool to Dublin and return by a different steamer?
- Q2:** Four persons enter a railway carriage in which there are six seats; in how many ways can they take their places ?
- Q3:** Out of 7 consonants and 4 vowels, how many words can be made each containing 3 consonants and 2 vowels?
- Q4:** If ${}^nC_{12} = {}^nC_8$, find ${}^nC_{17}$, ${}^{22}C_n$.
- Q5:** In how many ways can 16 things divided equally among 4 persons?
- Q6:** How many signals can be made by hoisting 6 differently coloured flags one above the other, when any number of them may be hoisted at once?
- Q7:** From 3 cocoa nuts, 4 apples and 2 oranges, how many selections of fruit can be made, taking at least one of each kind ?
- Q8:** There are p points in space, no four of which are in the same plane with the exception of q , which are all in the same straight line; find the number (1) of straight lines, (2) of triangles which result from joining them.
- Q9:** Find the sum of all numbers greater than 10000 formed by using the digits 13579, no digit repeated in any number.
- Q10:** Show that the number of permutations which can be formed from $2n$ letters which are either a 's or b 's is greatest when the number of a 's is equal to the number of b 's.